THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 10: **How Quantities Combine (Understanding Data)** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The average of 12 numbers is 89. When a number 63 is included, the average of all 13 numbers becomes: 1

(a) 89 **(b)** 88 **(c)** 87 **(d)** 86

2. The average of 27 numbers is 84. If the number 58 is removed, the average of the remaining 26 numbers is: 1

(a) 87 **(b)** 84 **(c)** 86 **(d)** 85

3. The average of 40 numbers is 73. The average of all 60 numbers (including 20 more numbers) is 70. The average of the 20 additional numbers is: 1

(a) 64 **(b)** 66 **(c)** 65 **(d)** 63

4. A group of 12 people has average weight 71 kg. Another group has average weight 74 kg. Together the average weight is 73 kg. The number of people in the second group is: 1

(a) 24 **(b)** 23 **(c)** 25 **(d)** 26

5. 360 mL of a 35% sugar solution is mixed with 680 mL of a 9% sugar solution. The concentration of the mixture is: 1

(a) 19% **(b)** 18% **(c)** 20% **(d)** 17%

6. Marks of 62, 78 and 79 are given weights 3, 4 and 4 respectively. The weighted average is: 1

(a) 74 **(b)** 73 **(c)** 76 **(d)** 75

7. The average monthly salary of 14 employees is ₹56 thousand. An employee earning ₹28 thousand leaves and a new employee joins, and the average rises by ₹3 thousand. The new employee's salary (in ₹ thousand) is: 1

(a) 70 **(b)** 75 **(c)** 67 **(d)** 73

8. A shop earns a profit of 9% on sales of ₹3000 and 24% on sales of ₹2000. The overall profit percentage on total sales is: 1

(a) 15% **(b)** 17% **(c)** 16% **(d)** 14%

9. The average of 37 numbers is 86. When a number 48 is included, the average of all 38 numbers becomes: 1

(a) 86 **(b)** 84 **(c)** 87 **(d)** 85

10. The average of 31 numbers is 66. If the number 156 is removed, the average of the remaining 30 numbers is: 1

(a) 63 **(b)** 65 **(c)** 64 **(d)** 62

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. The mean of 32 observations is 44. One observation 106 is removed. Find the mean of the remaining observations. 2

12. The average monthly salary of 18 employees is ₹48 thousand. An employee earning ₹33 thousand leaves and a new employee joins; the average rises by ₹2 thousand. Find the new employee's monthly salary. 2

13. A cyclist rides from home to a market at 42 km/h and returns by the same route at 56 km/h. Find the average speed for the whole trip. 2

14. The mean of 36 observations is 56. One observation 126 is removed. Find the mean of the remaining observations. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In a class, 21 students have an average of 56 marks and the remaining students have an average of 36 marks. The class average is 41. Find the number of remaining students. 3

16. The average score of 45 students in a test is 80. When 54 more students take the test, the average score of all 99 students becomes 68. Find the average score of the 54 additional students. 3

17. 180 mL of a 36% acid solution is diluted by adding 60 mL of water. Find the new concentration. 3

OR

In a class, 38 students have an average of 47 marks and the remaining students have an average of 37 marks. The class average is 42. Find the number of remaining students. 3

18. The average score of 32 students in a test is 74. When 40 more students take the test, the average score of all 72 students becomes 84. Find the average score of the 40 additional students. 3

OR

160 mL of a 11% acid solution is diluted by adding 60 mL of water. Find the new concentration. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Three sections have 23, 25 and 30 students with average marks 82, 88 and 64 respectively. Find the average marks of all the students. 5

OR

510 mL of a 22% sugar solution is mixed with 680 mL of a 15% sugar solution. Find the percentage concentration of the mixture. 5

20. Three sections have 30, 30 and 36 students with average marks 61, 41 and 83 respectively. Find the average marks of all the students. 5

OR

320 mL of a 34% sugar solution is mixed with 400 mL of a 52% sugar solution. Find the percentage concentration of the mixture. 5

END OF QUESTION PAPER — Total: 40 Marks